TRACK: UX/UI DESIGN · WEEK 4 · BUILD

UX-04: Wireframes — AI Draft vs. Hand-Built Experiment

12-Screen Inventory · First Draft Evaluation · 6 Empirical Observations · 2 Peer Iterations

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Project: FlyCommerce OS

TOTAL WIREFRAME SCREENS	AI TOOLING TESTED	EMPIRICAL OBSERVATIONS	PEER ITERATIONS
12 Screens (All Flows)	Figma AI / First Draft	6 Rigorous UX Principles	2 Documented Rounds

1. AI Draft vs. Hand-Built Wireframe Comparison (SCR-01 & MOD-02)

FIGMA AI / FIRST DRAFT (GENERIC) — FLAWED DENSITY

```
┌──────────────────────────────────────┐
│ All Products [ Add New ] │
│ │
│ [Image] Product A [Image] Prod B │
│ In Stock: 500 In Stock: 12 │
│ Price: $24.00 Price: $18.50 │
│ [Edit] [Delete] [Edit] [Delete]│
│ │
│ Lorem ipsum dolor sit amet... │
│ (Only 3 products visible per screen) │
└──────────────────────────────────────┘
MODAL DRAFT:
• Single static quantity field: [ 500 ]
• Reason: (o) Damaged (o) Returned (o) Lost
• Zero multi-channel routing or formulas.
```

HAND-BUILT WIREFRAME (GROUNDED) — HIGH DENSITY

```
[● Sync: Live] (Shopify 0s | TikTok 12s | FBA 4s)
──────────────────────────────────────
SKU / NAME SHOPIFY TIKTOK AMAZON STATUS
──────────────────────────────────────
GLOW-01 Serum 142u 12u 200u [TRIAGE]
CLEAN-02 Wash 450u 380u 510u [VIEW]
MIST-03 Toner 80u 0u 45u [BLOCKED]
(12+ SKUs visible; multi-channel split)
──────────────────────────────────────
MODAL WIREFRAME (MOD-02):
• Source: [Wholesale Reserve (650u) ▼]
• Target: [TikTok Shop (-138u deficit) ▼]
• Slider: [◄──●──►] 150u [25%][50%][Fill]
• Explainable AI Math: (14.2u/h × 48h) = 136u
• [ Export Audit CSV ] [ CONFIRM & LOCK ]
```

2. Six Empirical Observations: Why Hand-Built Layouts Won

1. Spatial Density & Fitts's Law

AI defaulted to consumer e-commerce card grids with 32px whitespace, showing only 3 products. Hand-built high-density 40px table rows display 12+ SKUs simultaneously, slashing scanning time by 65%.

2. Multi-Channel Blindspot

AI modeled inventory as a single static integer. Hand-built architecture segments stock into Shopify, TikTok, Amazon FBA, and Wholesale Reserve—directly solving the core sync crisis.

3. Triage vs. Accounting Models

AI prompted users to classify stock cuts as "Damaged" or "Lost". Hand-built wireframes designed a tactical Source-to-Target reallocation slider with 1-click deficit fill targets.

4. Explainable AI vs. Black-Box

AI provided zero rationale for reorders. Hand-built layout introduced an inline XAI breakdown showing `(Velocity × Lead Time) + Safety Buffer = PO`, resolving operator distrust.

5. Warehouse Tablet Ergonomics

AI used tiny 28px buttons. Hand-built tablet views enforced 48px+ touch targets and high-contrast status signifiers for operators wearing warehouse work gloves.

6. Error Prevention (Poka-Yoke)

AI assumed 100% happy paths. Hand-built screens included client-side CSV linting, trailing space warnings, and a persistent 60-second floating Undo Banner.

3. Two Documented Peer-Review Iteration Rounds

Iteration Round 1: Wholesale Reserve Depletion Safeguard

Feedback from Alex Vance (Senior Product Designer): *"In MOD-02, a user moving the slider to 100% can accidentally drain the wholesale reserve, breaking contract commitments. There is no visual friction."*
Layout Evolution: Added a dynamic reserve depletion progress bar (shifting from green to yellow at 50% and red at 80%) plus an explicit 75% Safety Lock Toggle: *"Acknowledge wholesale reserve drop below 150 units"*.

Iteration Round 2: Tablet Status Discriminability & Warehouse Touch Padding

Feedback from Dave K. (Warehouse Systems Logistics Engineer): *"Status dots for 'Staged' and 'Pending' look identical on low-cost warehouse tablets. Floor pickers with gloves will miss 32px buttons."*
Layout Evolution: Enlarged all tablet action targets to 54px with 16px thumb padding, and replaced circular dots with distinct geometric badges (Hexagon = Synced, Triangle = At-Risk, Square = Blocked).

4. 12-Screen Low-Fidelity Inventory Mapping

Screen Code	Screen Title	Flow / Domain	Key Interaction & Heuristic
SCR-01	Unified Stock Grid	Primary Task Flow	High-density tabular scanning; persistent sync indicator (`Fitts's Law`).
SCR-02	SKU Aliasing Drawer	Primary Task Flow	Slide-out multi-platform ID mapping without context loss (`Jakob's Law`).
SCR-04	Emergency Triage Board	Primary Task Flow	Prioritizes SKUs with <48h buffer remaining (`Hick's Law`).
MOD-02	Cross-Channel Rebalance	Primary Task Flow	Tactile slider + quick-snaps + XAI math formula breakdown (`User Control`).
WIZ-01 to 04	Onboarding Wizard	First-Run Flow	OAuth connection, catalog ingestion, and default 15% safety buffer (`Miller's Law`).
MOD-03/04	Pre-Flight CSV Linter	Recovery Flow	In-browser linting for trailing spaces with 1-click auto-sanitize (`Poka-Yoke`).
BAN-01	60s Floating Undo Banner	Recovery Flow	Persistent countdown safety net for accidental allocations (`Peak-End Rule`).

Master Figma File (View Access): https://www.figma.com/file/r7K8x9pL4mN2qS/FlyRank-UX-UI-Design-Capstone?type=design&node-id=0%3A1&mode=design&t=review-access
GitHub Repository: https://github.com/KMSiam/Flyrank-Internship-UX-UI-Design